Hypothesis Testing

Hypothesis testing is a method used to decide whether sample data provides enough evidence to support or reject a claim about a population.
Think of it as a formal way of answering:
"Is this observed result just random chance, or is something real happening?"

Example: Battery Life Claim

Suppose a company claims:
"Our batteries last 10 hours on average."
You test 100 batteries and find:

- Sample mean = 9.2 hours

Now you want to know:
Is the difference (9.2 vs 10) large enough to conclude the claim is false?
This is a hypothesis testing problem.

Step 1: State the Hypotheses

Null Hypothesis (H0H_0H0)

The default assumption.
H0:μ=10H_0: \mu = 10H0:μ=10
Meaning:
The average battery life is 10 hours.

Alternative Hypothesis (H1H_1H1)

The claim you're investigating.
H1:μ≠10H_1: \mu \ne 10H1:μ□=10
Meaning:
The average battery life is not 10 hours.

Step 2: Assume the Null Hypothesis Is True

This is important.
We start by pretending the company's claim is correct.
Then we ask:
If the true mean really is 10 hours, how likely is it to observe a sample mean as extreme as 9.2?

Step 3: Calculate a Test Statistic

The most common is the z-statistic:
z=x¯−μSEz=\frac{\bar{x}-\mu}{SE}z=SEx¯−μ
where:

- x¯\bar{x}x¯ = sample mean
- μ\muμ = hypothesized mean
- SESESE = standard error

The z-score measures how many standard errors the sample mean is away from the claimed mean.
z=x−μσz = \frac{x - \mu}{\sigma}z=σx−μ
z=(1.2)−(0)1=1.2z=\frac{(1.2)-(0)}{1}=1.2z=1(1.2)−(0)=1.2
Φ(z)≈88.5%\Phi(z)\approx 88.5\%Φ(z)≈88.5%
xxx

1.

μ\muμ

0.

σ\sigmaσ

1.

Step 4: Find the p-value

The **p-value** answers:
Assuming H0H_0H0 is true, what is the probability of obtaining a result at least this extreme?
Small p-value:

- Result is unlikely under H0H_0H0
- Evidence against H0H_0H0

Large p-value:

- Result is plausible under H0H_0H0
- Not enough evidence against H0H_0H0

Step 5: Compare With Significance Level

Usually:
α=0.05\alpha = 0.05α=0.05
(5% significance level)
Decision rule:

- If p<0.05p < 0.05p<0.05, reject H0H_0H0
- If p≥0.05p \ge 0.05p≥0.05, fail to reject H0H_0H0

Example

Suppose:

- Claimed mean = 10
- Sample mean = 9.2
- Standard error = 0.25

Then:
z=9.2−100.25=−3.2z=\frac{9.2-10}{0.25} =-3.2z=0.259.2−10=−3.2
A z-score of -3.2 is quite far from 0.
The corresponding p-value is approximately:
p≈0.0014p \approx 0.0014p≈0.0014
Since:
0.0014<0.050.0014 < 0.050.0014<0.05
we reject H0H_0H0.
Conclusion:
There is strong evidence that the average battery life is not 10 hours.

Connection to Confidence Intervals

Confidence intervals and hypothesis tests are closely related.
Suppose a 95% confidence interval is:
(68.04, 71.96)(68.04,\;71.96)(68.04,71.96)
If someone claims:
μ=75\mu = 75μ=75
Since 75 is outside the interval:

- Reject H0:μ=75H_0: \mu = 75H0:μ=75
- at the 5% significance level

If the claimed value is inside the interval:

- Fail to reject H0H_0H0

Possible Outcomes

Correct Decisions

Reality	Decision
H0H_0H0 true	Fail to reject
H0H_0H0 false	Reject

Errors

Type I Error

Reject a true null hypothesis.
Example:

- Drug actually works.
- Study incorrectly concludes it doesn't.

Probability:
P(Type I Error)=αP(\text{Type I Error})=\alphaP(Type I Error)=α
Usually 5%.

Type II Error

Fail to reject a false null hypothesis.
Example:

- Drug doesn't work.
- Study concludes there's insufficient evidence against it.

Real-Life Examples

Medical Research

Claim:
New drug lowers blood pressure.
Hypothesis test determines whether observed improvement is statistically significant.

A/B Testing

Claim:
New website design increases conversions.
Hypothesis test compares conversion rates.

Manufacturing

Claim:
Average product weight is 500 g.
Samples are tested to ensure quality standards.

Election Polls

Claim:
Candidate A is leading Candidate B.
Hypothesis testing helps determine whether the observed lead is likely real or due to sampling variation.

The Big Picture

Hypothesis testing is essentially:

1. Make a claim (H0H_0H0)

2. Assume it's true
3. Measure how unusual your sample is under that assumption
4. Use probability to decide whether the sample is too unusual to be explained by chance

That's why probability is the foundation of inferential statistics: hypothesis testing is fundamentally a probability-based decision process.